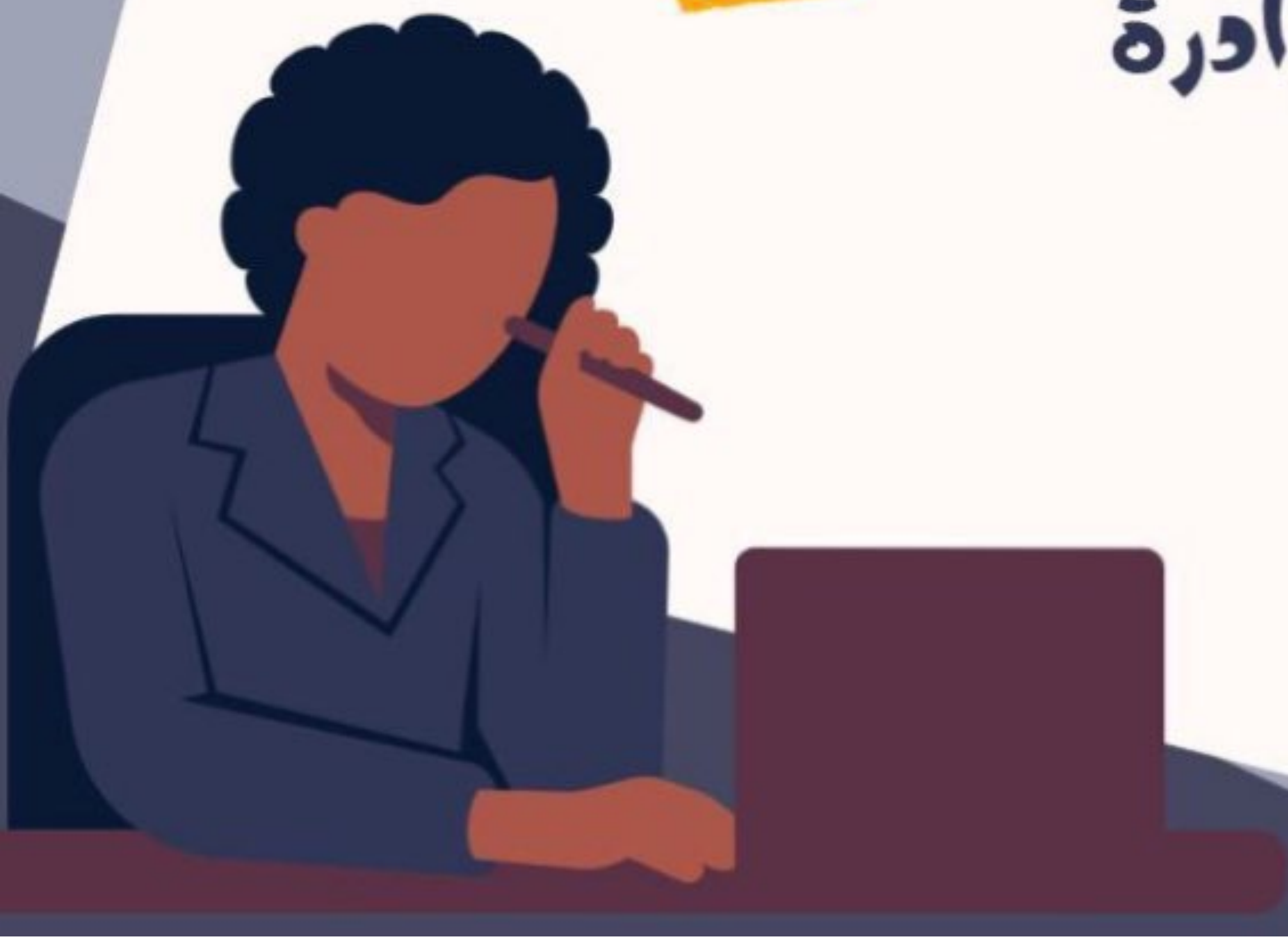
لاب تحليلية

مكمل مادة الفاينل

إعداد: شيماء غوادرة



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1. The most correct statement of the following is:

1) Oxidation is defined as the gain of electrons.

2) Secondary standard titrants should be standardized before use.

3) The purity of soda ash is measured as % Na_2CO_3

- All of them
- None of them
- 2&3
- 1&3

2. Back titration is more useful than normal titration when:

1) The titration reaction is slow

2) One of the products is volatile

3) Both acid and base are soluble

4) the titration involve a strong acid-strong base reaction

- 3&4
- 2&4
- 2&3
- 1&2

3. A sample of 0.4910 g of the antacid tablet was dissolved in 25.00 mL of 0.1006 M solution of HCl. After heating, the solution was back titrated with 14.71 mL of 0.09913 M solution of NaOH. The% $CaCO_3$ (100 g/mol) in the tablet is:

- 43.05
- 31.20
- 21.52
- 10.76

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4. For the following compounds, The light sensitive titrants are:

1) Na_2CO_3 2) $AgNO_3$ 3) $KMnO_4$ 4) $NaOH$

5) K_2CrO_4 6) KCl:

- 2&4
- 2&3
- 1&3
- 4&6

5. A sample of 25.0 ml of water was titrated with 5.40 ml of 0.0912 M HCl to reach end point. The total alkalinity measured as part per million $CaCO_3$ (100 g/mol) is:

- 115
- 492
- 242
- 985

6. A sample of 10.0 ml of water was titrated with 6.60 ml of 0.0103 M $AgNO_3$ to reach end point. The chloride content measured as part per million Cl^- (35.5 g/mol) is:

- 241
- 492
- 48.3
- 96.5

7. The molar solubility of Ag_2CrO_4 ($K_{sp}=1.2x10^{-12}$) is( )than AgCl ($K_{sp}=1.0* 10^{-10}$)and so precipitate ( ).

- higher, first
- lower, first
- higher, last

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8. In gravimetric analysis, the ideal product should be:

- Impure, insoluble and easily filterable
- Very pure, soluble and easily filterable
- Very pure, insoluble and less filterable
- Very pure, insoluble and easily filterable

9. Back titration is more useful than normal titration when:

1) The titration reaction is slow
2) One of the products is volatile
3) Both acid and base are soluble
4) the titration involve a strong acid-strong base reaction

- 3&4
- 2&4
- 2&3
- 1&2

10. State F for False statements and T for True statements of the following:

1) The major component of vinegar is acetic acid.

2) The total alkalinity of water is measured as part per million $BaSO_4$

3) K_2CrO_4 is the indicator used in the standardization of $KMnO_4$

- T, T, T
- T, T, F
- F, F, F
- T, F, F

Group

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11. A sample of 25.0 ml of water was titrated with 6.60 ml of 0.0103 M $AgNO_3$ to reach the end point. The chloride content measured as part per million Cl^- (35.5 g/mol) is:

- 242
- 492
- 48.3
- 96.5

12. A sample of 0.555 g of unknown containing $CaCl_2$ (111 g/mol) was dissolved in distilled water and titrated with 0.0500M $AgNO_3$ requiring 39.0 ml to reach the end point. The % $CaCl_2$ in the sample is:

- 14.1
- 39.0
- 19.5
- 78.0

13. A sample of 0.500 g of unknown containing only NaCl and $CaCl_2$ was dissolved and the Ca^{+2} in the solution is precipitated as $CaCO_2$ (100 g/mol) producing 0.215 g. The mass percent of NaCl (58.5 g/mol) in the sample is:

- 79.6
- 47.7
- 52.3
- 14.1

Group

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14· The indicator used in the titration of $KMnO_4$ with CO_2 is:

- $KMnO_4$
- Na_2CO_3
- $AgNO_3$
- K_2CrO_4

15· A sample of 0·135 g of $Na_2C_2O_4$ (134 g/mol) was dissolved and titrated with KMnO4 requiring 11·3 ml to reach the end point·

$2MnO_4^- + 5C_2O_4^{-2} + 16H^+ \longrightarrow 2Mn^{+2} + 10CO_2$

The molar concentration of $KMnO_4$:

- 0·0446
- 0·0357
- 0·111
- 0·0178

16· A sample of 0·9709g sodium carbonate is dissolved in water and diluted to 100mL· A sample of 25·00mL is taken and titrated to the equivalence point with 16·90mL of a 0·1022M HCl solution· The percentage of Na_2CO_3 (106 g/mol) in the sample:

$CO_3^{-2} + 2H^+ \longrightarrow H_2CO_3$

- 75·42%
- 37·71%
- 18·8%
- 97·09%

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17. The solubility product constants (Ksp) of AgCl and Ag_2CrO_4 precipitates respectively are 1.0×10^{-10} and 1.2×10^{-12}.
The correct statement of the following is:

- AgCl less soluble than Ag_2CrO_4 and precipitate last
- Ag_2CrO_4 less soluble than AgCl and precipitate first
- AgCl more soluble than Ag_2CrO_4 and precipitate first
- Ag_2CrO_4 more soluble than AgCl and precipitate last

18. A mixture of 100 mg of $CaCl_2$ (111 g/mole) and 150 mg of NaCl (58.45 g/mole) was dissolved and diluted to 500 ml, the part per million of chloride (35.45 g/mole) in the solution is:

- 155
- 246
- 310
- 123

19. A sample of 0.1228 g of the antacid tablet was dissolved in 25.00 mL of 0.1006 M solution of HCl. After heating, the solution was back titrated with 14.71 mL of 0.09913 M solution of NaOH. The % $CaCO_3$ (100g/mol) in the tablet is:

- 10.76
- 21.52
- 43.03
- 31.20

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20. A sample of 0.500 g of a mixture that may contain carbonate, bicarbonate, or hydroxide dissolved and titrated with 0.123 M HCl. The pH curve illustrates that the two end points are respectively at 18.2 and 30.1 ml. The mass percent of Na_2CO_3 (106 g/mol) in the sample is:

- 31.0
- 47.5
- 16.4
- 78.5

H2O Group

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